

STATISTICAL LEARNING THEORY (03/03/26)

Summer Term 2026

Other course names:

Machine Learning (Data Science Master)
Adv. Machine Learning (AIS Master)

MOTIVATION

Example:

	$\in \mathbb{R}$ Weight (in g)	$\in \mathbb{R}$ Color (in $[0,1]$)	$\in \{0,1\}$ Tasty?
Papaya 1	800	0.1	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$
Papaya N	1200	0.7	1

0 for TASTY? means "not tasty" and 1 means "tasty".

Let's call the information in this table our training data.

Based on that, we try to find

$$h: \mathbb{R} \times \mathbb{R} \rightarrow \{0, 1\}$$

i.e., a function that takes a tuple of (weight, color) and outputs a prediction of whether a papaya is tasty (1) or not (0).

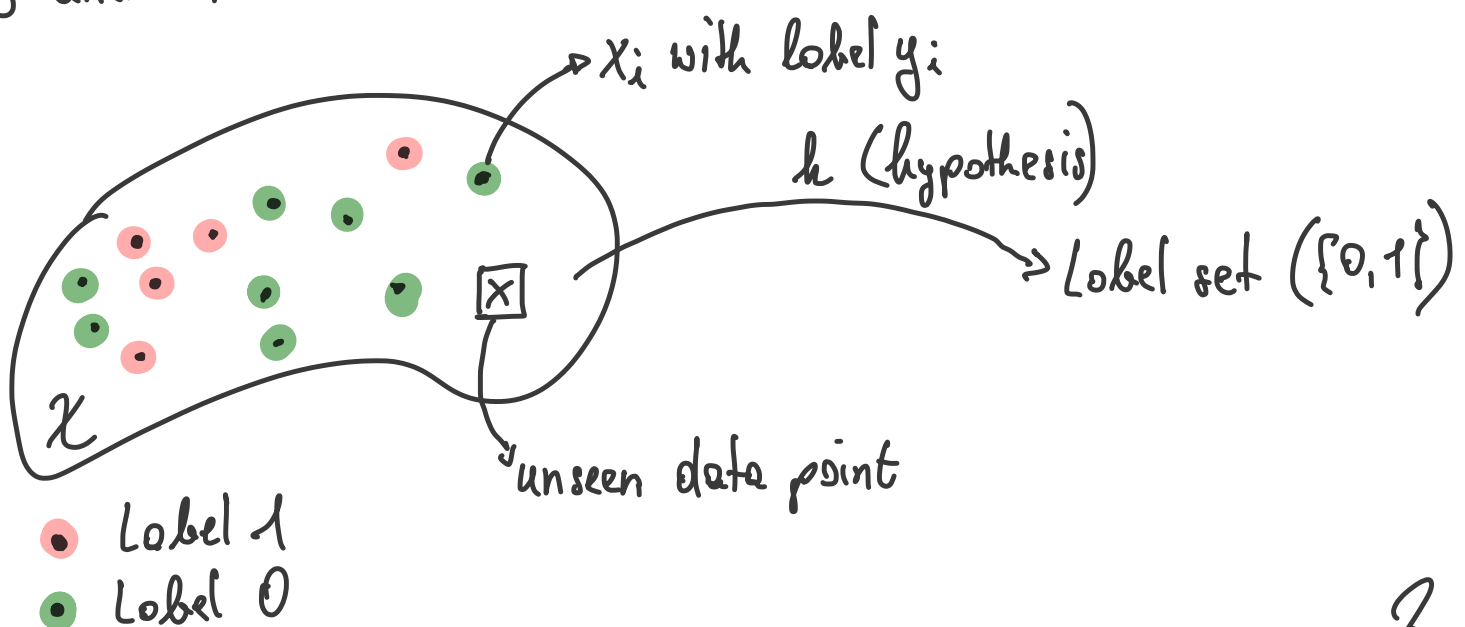
We are going to call such a function (h) a hypothesis.

More formally, the available data comes in the form of

$$((x_1, y_1), \dots, (x_N, y_N)) = S$$

with (in our example) $x_i \in \mathbb{R}^2$ and $y_i \in \{0, 1\}$.
domain $\mathcal{X}$ Label set $\mathcal{Y}$

We say a learner (some algorithm) receives the training data S and returns h !



Two assumptions we are going to make initially:

A.1. All the x_i 's are drawn independently and identically distributed (*iid*) from some (unknown) distribution D over our domain X .

A.2. The x_i 's are labeled by some (unknown) function

$$f: X \rightarrow Y$$

called the *true labeling function*.

This means our training data is of the form

$$S = \left(\underbrace{(x_1, f(x_1))}_{y_1}, \dots, \underbrace{(x_N, f(x_N))}_{y_N} \right)$$

What do we care about?

$$\left\{ x \in X : \underbrace{h(x)}_{\text{hypothesis}} \neq \underbrace{f(x)}_{\text{true labeling function}} \right\} = A$$

i.e., all points in our domain X for which h differs from f .

Specifically, we are interested in the probability of making a mistake

$$(h(x) \neq f(x)) \quad \mathcal{D}(A) = \mathcal{D}\left(\{x \in X : h(x) \neq f(x)\}\right) = \mathbb{P}_{x \sim D}[h(x) \neq f(x)]$$

$$(*) \quad \boxed{=: L_{D,f}(h)} \quad \text{generalization error}$$

The empirical version of (x) is

$$\frac{1}{N} \cdot \left| \left\{ i \in \{1, \dots, N\} : h(x_i) \neq f(x_i) \right\} \right| =: \boxed{L_S(h)}$$

We call this the **empirical error/risk**.

$$\left(\begin{array}{l} \text{Notation: } [N] = \{1, \dots, N\} \\ S|_x = (x_1, \dots, x_N) \end{array} \right)$$

We can also write the empirical error as

$$L_S(h) = \frac{1}{N} \cdot \sum_{i=1}^N \mathbb{1}_{h(x_i) \neq f(x_i)} \quad \text{with}$$

$$\mathbb{1}_{h(x) \neq f(x)} = \begin{cases} 1, & \text{if } h(x) \neq f(x) \\ 0, & \text{else} \end{cases}$$

Claim (relating L_S to $L_{D,f}$):

$$\mathbb{E}_{S|_x \sim D^N} [L_S(h)] = L_{D,f}(h)$$

Proof: next week!

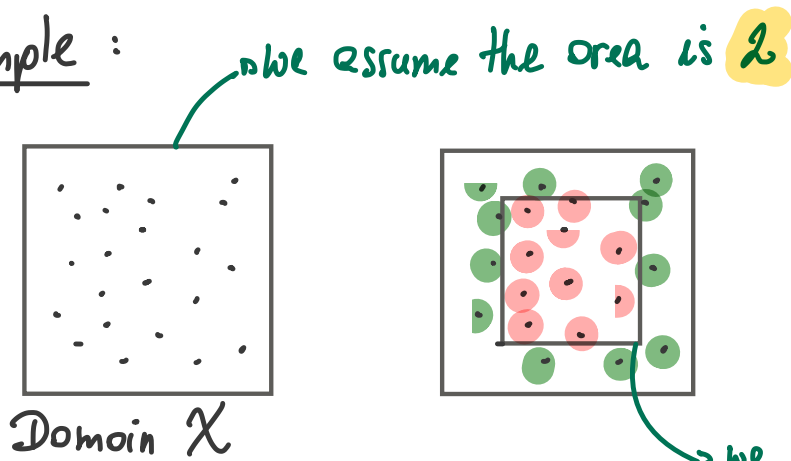
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$$\begin{aligned}
\mathbb{E}_{S|x \sim \mathcal{D}^m} [L_S(h)] &= \mathbb{E}_{S|x \sim \mathcal{D}^m} \left[\frac{1}{N} \cdot \sum_{i=1}^N \mathbb{1}_{h(x_i) \neq f(x_i)} \right] \quad // \text{by def.} \\
&= \frac{1}{N} \cdot \sum_{i=1}^N \mathbb{E}_{x_i \sim \mathcal{D}} [\mathbb{1}_{h(x_i) \neq f(x_i)}] \quad // \text{by linearity of } \mathbb{E}[\cdot] \\
&= \frac{1}{N} \cdot \sum_{i=1}^N \mathbb{E}_{x \sim \mathcal{D}} [\underbrace{\mathbb{1}_{h(x) \neq f(x)}}_{\text{only takes values 0 or 1}}] \quad // \text{as all } x_i \text{'s are iid} \\
&= \frac{1}{N} \cdot \sum_{i=1}^N \mathbb{P}_{x \sim \mathcal{D}} [h(x) \neq f(x)] \\
&= \frac{1}{N} \cdot \mathbb{P}_{x \sim \mathcal{D}} [h(x) \neq f(x)] \\
&= \mathbb{P}_{x \sim \mathcal{D}} [h(x) \neq f(x)] \\
&= L_{\mathcal{D}, f}(h) \quad // \text{by def.}
\end{aligned}$$

Our first learning paradigm

Empirical risk minimization (ERM): As a learner only has access to S (our training data), it is "natural" to try to select h (our hypothesis) such that the empirical risk/error $L_S(h)$ is minimized. We call such a hypothesis an **empirical risk minimizer** (h_S)!

Example :



• label 0 } given by f
• label 1 }

The distribution on X is a uniform distribution.

Lets say we have an algorithm that returns h_S

$$h_S(x) = \begin{cases} y_i, & \text{if } \exists i \in \{1, \dots, N\} : x_i = x \\ 0, & \text{else} \end{cases}$$

Clearly, h_S is correct on all training data points in $S \rightarrow h_S$ is an empirical risk minimizer.

$$L_S(h_S) = 0!$$

But, on unseen data points drawn from D (which is a uniform distribution on X), h_S is only correct 50% of the time $\Rightarrow$

$$L_{D,f}(h_S) = \frac{1}{2}!$$

That's what we call overfitting!!!

Hypothesis class (H): We restrict searching for h to H , i.e., a class of functions from $X \rightarrow Y$ and we write

$$\text{ERM}_H(S) \in \underset{h \in H}{\text{argmin}} L_S(h)$$

ERM over finite hypothesis classes ($|H| < \infty$)

Assumption A.3: $\exists h^* \in H$ with $L_{D,f}(h^*) = 0!$
(realizability)

Now, any ERM hypothesis h_S will attain 0 empirical error (meaning $L_S(h_S) = 0!$), as h_S competes against h^* and h^* has $L_S(h^*) = 0$ (and also $L_{D,f}(h^*) = 0$ by A.3).

We know that $L_{D,f}(h_S) > \epsilon$, $\epsilon \in (0, 1)$, can only happen if our learner selects a hypothesis h_S with $L_S(h_S) = 0$, but $L_{D,f}(h_S) > \epsilon$.

We define

$$H_{\text{BAD}} = \{h \in H : L_{D,f}(h) > \epsilon\}$$

(set of BAD hypotheses)

We also define

$$M = \left\{ S|_X : \exists h \in H_{\text{BAD}}, L_S(h) = 0 \right\}$$

("misleading" training data sets)

Observation:

ERM hypothesis

$$\left\{ S|_X : L_{D,f}(h_S) \geq \epsilon \right\} \subseteq \left\{ S|_X : \exists h \in H_{\text{BAD}}, L_S(h) = 0 \right\} = M$$

Since

$$\left\{ S|_X : \exists h \in H_{\text{BAD}}, L_S(h) = 0 \right\} = \bigcup_{h \in H_{\text{BAD}}} \left\{ S|_X : L_S(h) = 0 \right\}$$

We have

$$\begin{aligned} \mathcal{D}^N \left(\left\{ S|_X : \exists h \in H_{\text{BAD}}, L_S(h) = 0 \right\} \right) &= \mathcal{D}^N \left(\bigcup_{h \in H_{\text{BAD}}} \left\{ S|_X : L_S(h) = 0 \right\} \right) \\ &\leq \sum_{h \in H_{\text{BAD}}} \mathcal{D}^N \left(\left\{ S|_X : L_S(h) = 0 \right\} \right) \end{aligned}$$

(by the union bound)

Let us fix one $h \in H_{\text{BAD}}$
for a moment!

$$\underbrace{\mathcal{D}^N \left(\{S|_X : L_S(h) = 0\} \right)}_{\text{one of the summation terms}} = \mathcal{D}^N \left(\{S|_X : \forall i \in \{1, \dots, N\} : h(x_i) = f(x_i)\} \right)$$

one of the summation terms

$$= \prod_{i=1}^N \mathcal{D} \left(\{x \in X : h(x) = f(x)\} \right)$$

(by our iid assumption A.1)

Remember that $h \in H_{\text{BAD}}$, so $L_{\mathcal{D}, f}(h) > \varepsilon$.

$$\begin{aligned} &= \prod_{i=1}^N \left(1 - \underbrace{L_{\mathcal{D}, f}(h)}_{> \varepsilon} \right) \\ &\leq \prod_{i=1}^N (1 - \varepsilon) \\ &= (1 - \varepsilon)^N \\ &\leq e^{-\varepsilon N} \quad (\text{next week}) \end{aligned}$$

Overall, we have

$$\begin{aligned}
 \mathbb{D}^N \left(\left\{ S|_X : L_{D,P}(h_S) > \varepsilon \right\} \right) &\leq \sum_{h \in H_{\text{BAD}}} \mathbb{D}^N \left(\left\{ S|_X : L_S(h) = 0 \right\} \right) \\
 &\stackrel{\text{ERM hyp.}}{\leq} \sum_{h \in H_{\text{BAD}}} e^{-\varepsilon N} \quad (\text{from before}) \\
 &\leq |H_{\text{BAD}}| \cdot e^{-\varepsilon N} \\
 &\leq |H| \cdot e^{-\varepsilon N} \quad \text{since } H_{\text{BAD}} \subseteq H
 \end{aligned}$$

If we want $|H| \cdot e^{-\varepsilon N}$ to be less than $\delta \in (0,1)$, we can solve for N , i.e.,

$$\begin{aligned}
 &|H| \cdot e^{-\varepsilon N} < \delta \\
 \Rightarrow &\boxed{N > \frac{1}{\varepsilon} \cdot \log \left(\frac{|H|}{\delta} \right)} \\
 &\quad \swarrow \text{error} \qquad \searrow \text{confidence } (1-\delta)
 \end{aligned}$$

This means, the probability of the generalization error of an ERM hypothesis h_S being larger or equal to $\varepsilon \in (0,1)$ is upper-bounded by $|H| \cdot e^{-\varepsilon N}$ (to be understood as the probability of a random draw of a dataset of size N , iid from the distribution $\mathcal{D}$).

Remark: MARKOV inequality

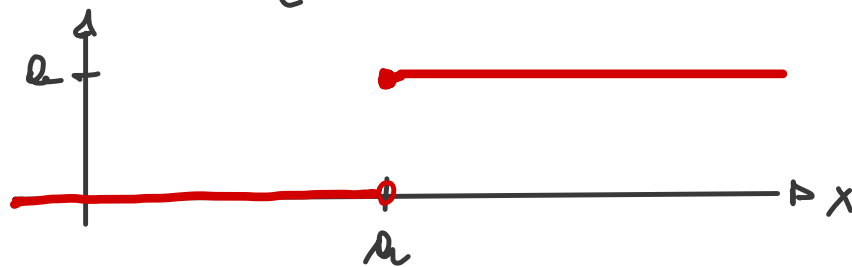
We have a probability space $(S, \mathcal{F}, \mathbb{D})$ and a non-negative random variable X ($X \geq 0$).

Claim: for all $a > 0$, it holds that

$$\mathbb{P}[X \geq a] \leq \frac{\mathbb{E}[X]}{a}$$

Proof: define $\varphi: S \rightarrow \mathbb{R}$

$$\varphi(x) = \begin{cases} a, & \text{if } x \geq a \\ 0, & \text{else} \end{cases}$$



We have that

$$0 \leq \varphi(X(x)) \leq X(x) \quad \text{for all } x \geq 0$$

$$\Rightarrow \int X d\mathbb{D} \geq \underbrace{\int \varphi(X(x)) d\mathbb{D}}_{a \cdot \mathbb{D}(\{w \in S: X(w) \geq a\})} \quad (\text{by monotonicity})$$

$$\text{Since } a > 0: \underbrace{\frac{1}{a} \cdot \int X d\mathbb{D}}_{\mathbb{E}[X]} \geq \mathbb{P}[X \geq a] \Rightarrow \mathbb{P}[X \geq a] \leq \frac{\mathbb{E}[X]}{a}$$

Corollary: Let $|H| < \infty$ (finite) and $\varepsilon, \delta \in (0, 1)$. Further, let N be an integer such that $N > \frac{1}{\varepsilon} \cdot \log\left(\frac{|H|}{\delta}\right)$. Then, for any labeling function $f: \mathcal{X} \rightarrow \{0, 1\}$ and any distribution $\mathcal{D}$ (for which realizability holds), we have that with probability of at least $1 - \delta$ over the choice of $S|_{\mathcal{X}}$ of size N , every ERM hypothesis h_S satisfies

$$L_{\mathcal{D}, f}(h_S) \leq \varepsilon.$$

Interpretation: For sufficiently large N , ERM_H returns a hypothesis h_S that is

PROBABLY APPROXIMATELY CORRECT (PAC).

Def. (PAC learnability): A hypothesis class H is PAC learnable if there exists a function $m_H: (0, 1)^2 \rightarrow \mathbb{N}$ and a learning algorithm A with the following properties: (I) for every $\varepsilon, \delta \in (0, 1)$ and (II) every distribution $\mathcal{D}$ over domain $\mathcal{X}$, and (III) every ^(true) labeling function $f: \mathcal{X} \rightarrow \{0, 1\}$, if (IV) realizability holds, then running A on $N > m_H(\varepsilon, \delta)$ iid instances from $\mathcal{D}$ (labeled by f) returns h_S such that with probability of at least $1 - \delta$ (over $S|_{\mathcal{X}}$), it holds that $L_{\mathcal{D}, f}(h_S) \leq \varepsilon$.

From now on, we will call our sample size m (not N).

Terminology: $m_H: (0,1)^2 \rightarrow \mathbb{N}$ is called the sample complexity (function).

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In our next step, we are going to remove the realizability assumption. (A.3). This means, there is no longer a $h^* \in H$ with $L_{D,f}(h^*) = 0$. So, the best possible thing we can hope for is a guarantee relative to $\min_{h \in H} L_{D,f}(h)$.
with realizability, this is 0!!!

Def. (Hoeffding inequality): Let $X_1, \dots, X_m$ be m iid random variables (RVs) taking values in $[a_i, b_i]$ for $i \in \{1, \dots, m\}$.

Then, given that

$$S_m = \sum_{i=1}^m X_i$$

it holds that

$$(a) \quad \mathbb{P}[S_m - \mathbb{E}[S_m] > \varepsilon] \leq e^{\frac{-2\varepsilon^2}{\sum_{i=1}^m (b_i - a_i)^2}}$$

$$(b) \quad \mathbb{P}[S_m - \mathbb{E}[S_m] < -\varepsilon] \leq \dots$$

Upon combination (using the union bound), we get

$$\mathbb{P}\left[\left|S_m - \mathbb{E}[S_m]\right| > \varepsilon\right] \leq 2 \cdot e^{-\frac{2\varepsilon^2}{\sum_i (b_i - a_i)^2}} \quad (\text{without proof!})$$

For us, the following alternative formulation will be helpful:

$$\mathbb{P}\left[\left|\frac{1}{m} \sum_{i=1}^m X_i - \mu\right| > \varepsilon\right] \leq 2 \cdot e^{-\frac{2\varepsilon^2 m}{(b-a)^2}}$$

with $\mu = \mathbb{E}[X_i]$ and $\mathbb{P}[a \leq X_i \leq b] = 1$ (for $i \in \{1, \dots, m\}$)

By using the Hoeffding inequality, we can already say the following: fix $\varepsilon > 0$; then, for a single hypothesis $h: X \rightarrow Y$, we have

$$\mathbb{P}_{S \sim D^m} \left[\left| \underbrace{L_S(h)}_{\frac{1}{m} \sum_{i=1}^m \mathbb{1}_{h(x_i) \neq f(x_i)}} - L_{D,f}(h) \right| > \varepsilon \right] \leq 2 \cdot e^{-2\varepsilon^2 m} \quad \left(\begin{array}{l} \text{since } b=1 \\ a=0 \end{array} \right)$$

Importantly, this only holds for a single h !!!

We actually want a bound that holds uniformly over all hypotheses in the class H . For finite hypothesis classes ($|H| < \infty$), we get such a result by the union bound.

Formally:

for a single h !

$$\mathbb{P}_{S|X \sim D^m} \left[\exists h \in H: |L_S(h) - L_{D,f}(h)| > \varepsilon \right] \stackrel{\text{u.B.}}{\leq} \sum_{h \in H} 2 \cdot e^{-2\varepsilon^2 m} \\ \leq |H| \cdot 2 \cdot e^{-2\varepsilon^2 m}$$

with realizability, we
had ε here not ε^2 !

We can next also get rid of the true labeling function f very easily by letting D be a distribution over X and Y , i.e., over the domain X and label set Y .

Let $Z := X \times Y$. We are going to adapt our definitions for the generalization and empirical error as follows:

$$L_D(h) = \mathbb{P}_{(x,y) \sim D} [h(x) \neq y] = D(\{(x,y) \in Z: h(x) \neq y\})$$

$$L_S(h) = \frac{1}{m} \cdot \left| \{i \in \{1, \dots, m\}: h(x_i) \neq y_i\} \right|$$

Def. (Agnostic PAC learnability): A hyp. class H of functions $h: X \rightarrow Y$ is **agnostic PAC learnable** if there exists $m_H: (0,1)^2 \rightarrow \mathbb{N}$ and an algorithm A with the following properties: (I) for every $\epsilon, \delta \in (0,1)$ and (II) every distribution $\mathcal{D}$ over $\mathcal{Z} := X \times Y$, when running A on m iid instances from $\mathcal{D}$ with $m \geq m_H(\epsilon, \delta)$, A returns a hypothesis h such that with probability of at least $1-\delta$ (over the choice of S) it holds that

$$L_{\mathcal{D}}(h) \leq \min_{h' \in H} L_{\mathcal{D}}(h') + \epsilon$$

Another generalization of setup:

Loss function $\ell: H \times X \times Y \rightarrow \mathbb{R}_+$

Example: "0-1 loss"

$$\ell^{0-1}(h, (x, y)) = \begin{cases} 1, & \text{if } h(x) \neq y \\ 0, & \text{if } h(x) = y \end{cases}$$

With this definition, we can write

$$\begin{aligned} L_{\mathcal{D}}(h) &= \mathbb{E}_{\substack{z \sim \mathcal{D} \\ (x,y)'}} [\ell^{0-1}(h, z)] = 1 \cdot \mathbb{P}[h(x) \neq y] + 0 \cdot \cancel{\mathbb{P}[h(x) = y]} \\ &= \mathbb{P}_{(x,y) \sim \mathcal{D}}[h(x) \neq y] \end{aligned}$$

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Uniform Convergence

Def. (ϵ -representative sample): A sample S is called ϵ -representative with respect to $Z = X \times Y$, hypothesis class H , loss function ℓ , and distribution D , if

$$\forall h \in H: |L_S(h) - L_D(h)| \leq \epsilon$$

Lemma: Assume that S is $\frac{\epsilon}{2}$ -representative wrt. Z, H, ℓ and D .

Then, any hypothesis h_S returned by

$$\text{ERH}_H(S) \in \underset{h' \in H}{\text{argmin}} L_S(h')$$

satisfies

$$L_D(h_S) \leq \min_{h \in H} L_D(h) + \epsilon$$

Proof: for any $h \in H$:

$$\begin{aligned} L_D(h_S) &\leq L_S(h_S) + \frac{\epsilon}{2} && // \text{by } \frac{\epsilon}{2}\text{-representativeness} \\ &\leq L_S(h) + \frac{\epsilon}{2} && // \text{since } h_S \text{ is an ERH hyp.} \\ &\leq L_D(h) + \frac{\epsilon}{2} + \frac{\epsilon}{2} && // \text{by } \frac{\epsilon}{2}\text{-representativeness} \\ &= L_D(h) + \epsilon && (*) \end{aligned}$$

We conclude: since $(*)$ holds for any $h \in H$, we get

$$L_D(h_S) \leq \min_{h \in H} L_D(h) + \epsilon$$

Def: (uniform convergence (UC)): A hypothesis class H has the UC property with respect to $Z = X \times Y$ and loss function L , if there exists $m_H^{uc} : (0,1)^2 \rightarrow \mathbb{N}$ such that for any $\epsilon, \delta \in (0,1)$ and every distribution D over $Z = X \times Y$, if S is an iid sample of size $m \geq m_H^{uc}(\epsilon, \delta)$ from D , then with probability of at least $1 - \delta$ (over the choice of S), S is ϵ -representative.

Corollary: If H satisfies the UC property with sample complexity function m_H^{uc} , then H is agnostic PAC learnable with

$$\underbrace{m_H(\epsilon, \delta)}_{\text{sample complexity for agnostic PAC}} \leq m_H^{uc}\left(\frac{\epsilon}{2}, \delta\right)$$

sample complexity
for agnostic PAC

by ERM (i.e., our learning paradigm).

Question: Is there a universal learner? By universal we mean a learner without any prior knowledge of the learning task/problem (given by D), but can be challenged by any task and still returns $A(S)$ with low $L_D(A(S))$.

Theorem (No-Free-Lunch): Let A be a learning algorithm for the task of binary classification with respect to the 0-1 loss, over domain X . Also let m be any number ($\in \mathbb{N}$) smaller than $|X|/2$ (representing the size of S). Then, there exists a distribution D over $X \times \underbrace{\{0,1\}}_Y$ such that

① $\exists f: X \rightarrow \{0,1\}$ with $L_D(f) = 0!$

and

② with probability of at least $1/4$ over the choice of $S \sim D^m$, we have

$$L_D(A(S)) \geq 1/p.$$

Interpretation: ① means that the task can be learned by another learner (with, e.g., $H = \{f\}$), and ② means that A fails on that task!

Corollary: Let $\mathcal{X}$ be infinitely large and H be the class of all functions from $\mathcal{X} \rightarrow \{0,1\}$, then H is not PAC learnable.

VAPNIK - CHERVONENKIS (VC) - Dimension

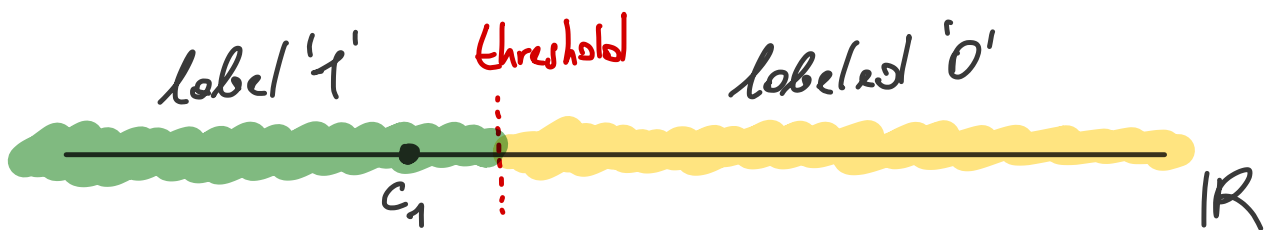
Def: Let H be a class of functions from $\mathcal{X} \rightarrow \{0,1\}$ and let $C \subset \mathcal{X}$, $C = \{c_1, \dots, c_m\}$. We define

$$H_C = \left\{ (h(c_1), h(c_2), \dots, h(c_m)) : h \in H \right\}$$

as the restriction of H to the subset C .

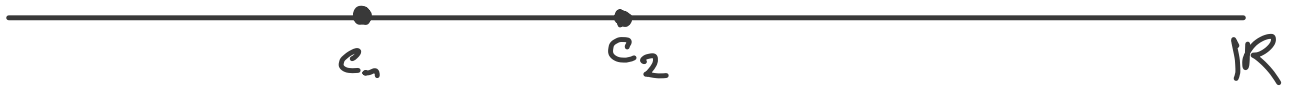
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Example: $C = \{c_1\}$, $c_1 \in \mathbb{R}$ and let H^{thr} be the class of thresholds on the real line $\mathbb{R}$.



$$H_C^{\text{thr}} = \{(1), (0)\} \Rightarrow |H_C^{\text{thr}}| = 2 = 2^1$$

Let's take $C = \{c_1, c_2\}$; without loss of generality
we say $c_1 \leq c_2$



We get

$$H_C^{\text{thr}} = \{(1,1), (1,0), (0,0)\}$$

$$\Rightarrow |H_C^{\text{thr}}| = 3$$

But, we would have $2^2 = 4$ possible labelings of $C = \{c_1, c_2\}$.

Def. (Shattering): A hypothesis class H shatters a finite set C of size m if $|H_C| = 2^m$.

Claim: The class H^{thr} on $\mathcal{X} = \mathbb{R}$ is PAC-learnable
with

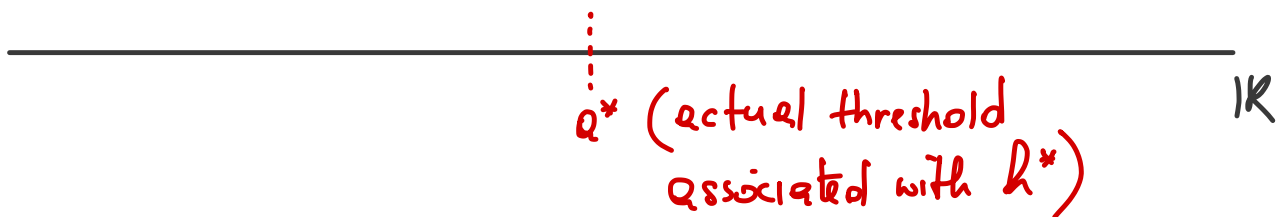
$$m_{H^{\text{thr}}}(\epsilon, \delta) \leq \left\lceil \frac{1}{\epsilon} \cdot \log\left(\frac{2}{\delta}\right) \right\rceil$$

$$H^{\text{thr}} = \{h_a : a \in \mathbb{R}\}, \quad h_a(x) = \begin{cases} 1, & \text{if } x < a \\ 0, & \text{else} \end{cases}$$

$\uparrow$
 threshold

$$\text{(or } h_a(x) = \mathbb{1}_{x < a}\text{)}$$

Proof: We assume realizability, i.e., $\exists h^* \in H^{\text{thr}}$ such that $L_{D,f}(h^*) = 0$.



We let $a_0 \in \mathbb{R}$ be such that

$$D(\{x \in \mathbb{R} : x \in (a_0, q^*)\}) = \varepsilon$$

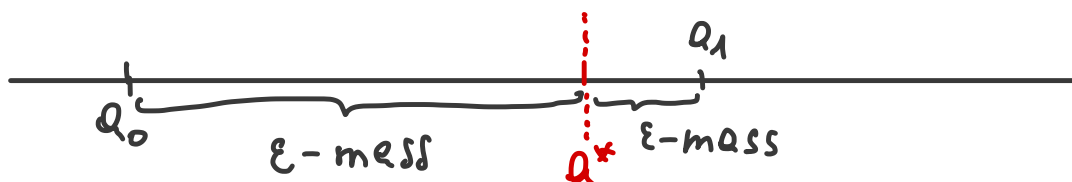
and let $a_1 \in \mathbb{R}$ be such that

$$D(\{x \in \mathbb{R} : x \in (q^*, a_1)\}) = \varepsilon$$

Special cases:

In case $D(\{x \in \mathbb{R} : x \in (a_0, q^*)\}) < \varepsilon$, then set $a_0 = -\infty$

In case $D(\{x \in \mathbb{R} : x \in (q^*, a_1)\}) < \varepsilon$, then set $a_1 = +\infty$



Next, we are going to define an ERM algorithm
(needed for PAC-learnability).



(1) Pick $b_0 \in \mathbb{R}$ and $b_1 \in \mathbb{R}$ from $S = ((x_1, y_1), \dots, (x_m, y_m))$

as

$$b_0 = \max \{x : (x, 1) \in S\}$$

$$b_1 = \min \{x : (x, 0) \in S\}$$

(2) Pick any threshold within (b_0, b_1) and call the corresponding hypothesis h_S .

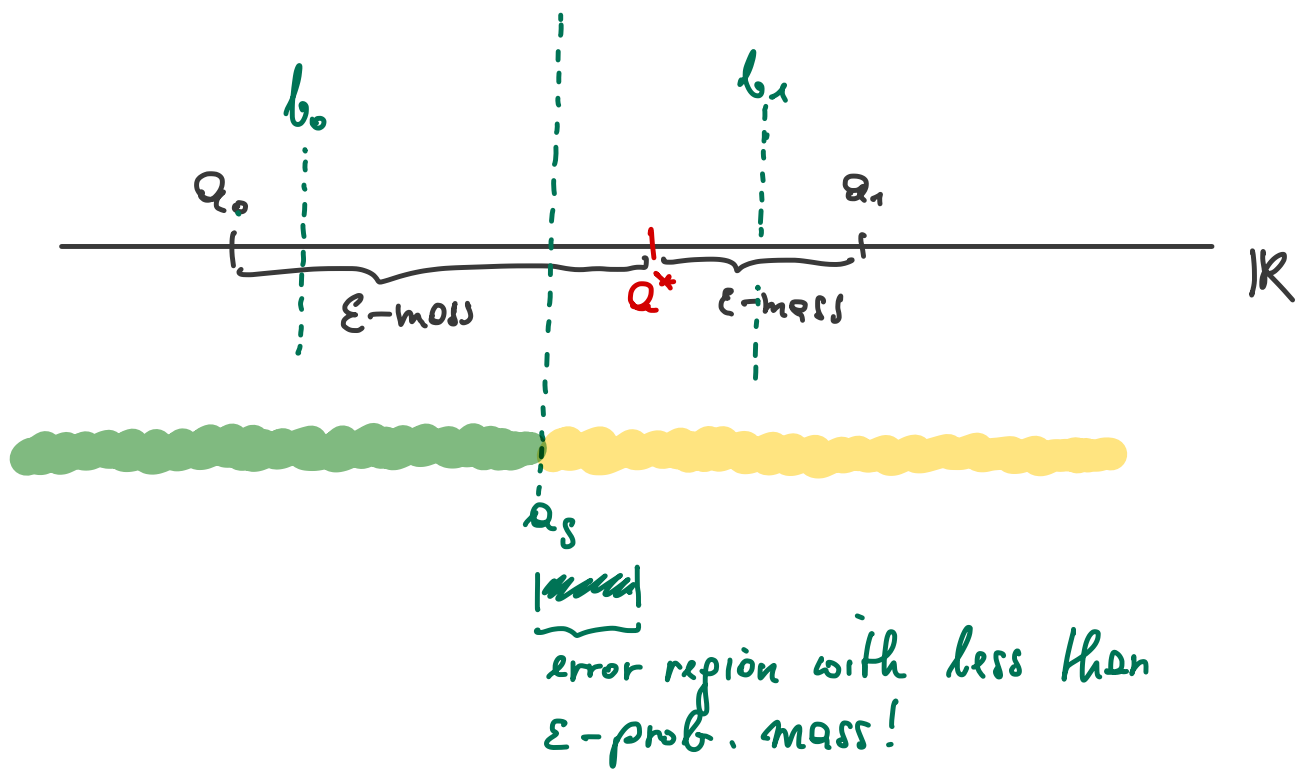
We see that h_S always has 0 empirical error !!!

By our construction of a_0 and a_1 , the claim is that for h_S (our ERM hypothesis) to have $L_{D,f}(h_S) \leq \epsilon$, it suffices that

$$(a) \quad b_1 \leq a_1 \quad \underline{\text{and}}$$

$$(b) \quad b_0 \geq a_0$$

(*)



Formally, we have

$$\mathbb{P}\left[L_{D,f}(h_S) > \epsilon\right] \leq \mathbb{P}\left[(b_0 < q_0) \text{ or } (b_1 > q_1)\right]$$

$\uparrow$
 $\text{ERh hyp.} \quad \leq \text{u.B.} \quad \mathbb{P}[b_0 < q_0] + \mathbb{P}[b_1 > q_1]$

To upper-bound $\mathbb{P}[L_{D,f}(h_S) > \epsilon]$, we need to check when $\underline{b_0 < q_0}$ (or $b_1 > q_1$). This only happens when there is no instance x in S that is labeled 1 such that $x \in (q_0, q^*)$.

We know that $\mathcal{D}(\{x \in \mathbb{R} : x \in (q_0, q^*)\}) = \epsilon$ (by construction)

Hence, not seeing an instance in (q_0, q^*) has probability of $1 - \epsilon$, and not seeing an instance in m iid samples has probability of $(1 - \epsilon)^m$.

Overall :
$$\mathbb{P}[L_{D,f}(h_S) > \varepsilon] \leq 2 \cdot (1-\varepsilon)^m$$

$$\leq 2e^{-\varepsilon m}$$

If we want $2e^{-\varepsilon m} < \delta$ for $\delta \in (0,1)$, we get

$$m > \frac{1}{\varepsilon} \cdot \log\left(\frac{2}{\delta}\right)$$

□

We have seen that finiteness of H is **sufficient**, but **not necessary** for PAC learnability.

05/05/26

Def. (VC-Dimension) : The VC-Dimension of H (i.e., a class of functions from $X \rightarrow \{0,1\}$), written as

$$VC(H)$$

is the maximal size of a set $C \subset X$ that is shattered by H .

Theorem: Let H be a class of functions from $X \rightarrow \{0, 1\}$.
if H has **infinite** VC-Dimension, then H is not PAC learnable.

Def. (growth function): Let H be a class of functions from $X \rightarrow \{0, 1\}$. The **growth function** of H , $\gamma_H: \mathbb{N}_0 \rightarrow \mathbb{N}_0$, is defined as

$$\gamma_H(m) = \max_{\substack{C \subset X \\ |C|=m}} |H_C| \quad \left| \begin{array}{l} \text{convention is} \\ \gamma_H(0) = 1 \end{array} \right.$$

(for H_{thr} on the real line
 $\gamma_{H_{\text{thr}}}(0) = 1, \gamma_{H_{\text{thr}}}(1) = 2, \gamma_{H_{\text{thr}}}(2) = 3, \dots$
 $(2^m - 2^{m-1} = 2)$)

Lemma (Sauer, Shelah, Peres, "Sauer's Lemma"): Let H be a class of functions from $X \rightarrow \{0, 1\}$ with $VC(H) = d < \infty$.

Then

$$\gamma_H(m) = 2^m \text{ if } m \leq d, \text{ BUT}$$

$$\gamma_H(m) \leq \left(\frac{e \cdot m}{d}\right)^d \text{ if } m > d$$

If we take any set C (of size m) and $\underbrace{|H| < \infty}_{\text{finite class}}$, we know

$$|H_C| \leq |H|$$

So, if $|H| < 2^m$, then obviously H can not shatter C of size m . This implies

$$VC(H) \leq \log_2(|H|)$$

Theorem: Let H be a class of functions from $X \rightarrow \{0, 1\}$ and $\ell: H \times X \times Y \rightarrow [0, c]$, $c > 0$, a loss function. For any distribution D over $X \times Y$ and $\delta \in (0, 1)$, we have with probability of at least $1 - \delta$ (over the choice of $S \sim D^m$)

$$\forall h \in H: |L_D(h) - L_S(h)| \leq c \cdot \sqrt{\frac{\mathcal{P} \cdot \log\left(\tilde{N}_H(2m) \cdot \frac{4}{\delta}\right)}{m}}$$

Proof: Take $S \sim \mathcal{D}^m$ and $S' \sim \mathcal{D}^m$.

lets look at $|L_{S'}(h) - L_S(h)|$. First, we expand the terms as follows:

$$|(L_{\mathcal{D}}(h) - L_S(h)) - (L_{\mathcal{D}}(h) - L_{S'}(h))| \geq |L_{\mathcal{D}}(h) - L_S(h)| - |L_{\mathcal{D}}(h) - L_{S'}(h)|$$

(reverse triangle ineq.)

if ① $|L_{\mathcal{D}}(h) - L_S(h)| > \varepsilon$ and

② $|L_{\mathcal{D}}(h) - L_{S'}(h)| < \frac{\varepsilon}{2}$, then $|L_{S'}(h) - L_S(h)| > \frac{\varepsilon}{2}$

Rewritten in terms of indicator functions:

$$\mathbb{1}_{|L_{\mathcal{D}}(h) - L_S(h)| > \varepsilon} \cdot \mathbb{1}_{|L_{\mathcal{D}}(h) - L_{S'}(h)| < \frac{\varepsilon}{2}} \leq \mathbb{1}_{|L_{S'}(h) - L_S(h)| > \frac{\varepsilon}{2}}$$

lets take the expectation over S' :

$$\mathbb{E}_{S'} \left[\mathbb{1}_{|L_{\mathcal{D}}(h) - L_{S'}(h)| < \frac{\varepsilon}{2}} \right] = \mathbb{P}_{S'} \left[|L_{\mathcal{D}}(h) - L_{S'}(h)| < \frac{\varepsilon}{2} \right]$$

by Hoeffding ineq. $\geq 1 - 2 \cdot e^{-\frac{2 \cdot (\frac{\varepsilon}{2})^2 m^2}{\sum_i (c-0)^2}}$

$$= 1 - 2e^{-\frac{\varepsilon^2 m}{2c^2}} \quad (\times)$$

$$\boxed{2} \quad \mathbb{E}_{S'} \left[\frac{1}{2} \mathbb{1}_{|L_{S'}(h) - L_S(h)| > \frac{\epsilon}{2}} \right] = \mathbb{P}_{S'} \left[|L_{S'}(h) - L_S(h)| > \frac{\epsilon}{2} \right]$$

Question: Can we further bound (x) $1 - 2e^{-\frac{\epsilon^2 m}{2c^2}}$

05/19/2026

1st lecture exam date = last lecture.

Proof continued:

→ if $m \geq 4c^2 \epsilon^{-2} \cdot \log(2)$ (we have to check this later)
we would get

$$\begin{aligned} 1 - 2e^{-\frac{\epsilon^2 m}{2c^2}} &\geq 1 - 2e^{-\frac{\epsilon^2 4c^2 \epsilon^{-2} \log(2)}{2c^2}} \\ &= 1 - 2e^{-2 \log(2)} \\ &= 1 - 2 \cdot \frac{1}{4} = \frac{1}{2} \end{aligned}$$

At the moment, we have

$$\frac{1}{2} \mathbb{1}_{|L_D(h) - L_S(h)| > \epsilon} \cdot \frac{1}{2} \leq \mathbb{E}_{S'} \left[\frac{1}{2} \mathbb{1}_{|L_{S'}(h) - L_S(h)| > \frac{\epsilon}{2}} \right]$$

$$\begin{aligned} \Leftrightarrow \frac{1}{2} \mathbb{1}_{|L_D(h) - L_S(h)| > \epsilon} &\leq 2 \cdot \mathbb{P}_{S'} \left[|L_{S'}(h) - L_S(h)| > \frac{\epsilon}{2} \right] \\ &\leq 2 \cdot \mathbb{P}_{S'} \left[\exists h \in \mathcal{H} : |L_{S'}(h) - L_S(h)| > \frac{\epsilon}{2} \right] \end{aligned}$$

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We can re-write this as

$$\frac{1}{m} \mathbb{1}_{\exists h \in H: |L_D(h) - L_S(h)| > \varepsilon} \leq 2 \cdot \mathbb{P}_{S'} \left[|L_{S'}(h) - L_S(h)| > \frac{\varepsilon}{2} \right]$$

(as $\frac{1}{m} \mathbb{1}_{|L_D(h) - L_S(h)| > \varepsilon}$ is either 0 or 1). So, if we now take the expectation with respect to S , we get

$$\mathbb{E}_S \left[\frac{1}{m} \mathbb{1}_{\exists h \in H: |L_D(h) - L_S(h)| > \varepsilon} \right] \leq 2 \cdot \mathbb{E}_{S, S'} \left[|L_{S'}(h) - L_S(h)| > \frac{\varepsilon}{2} \right]$$

Remember that $L_S(h) = \frac{1}{m} \cdot \sum_{i=1}^m \ell(h, z_i)$ and

$$L_{S'}(h) = \frac{1}{m} \cdot \sum_{i=1}^m \ell(h, z_i')$$

Hence,

$$|L_{S'}(h) - L_S(h)| = \frac{1}{m} \cdot \left| \sum_{i=1}^m (\ell(h, z_i') - \ell(h, z_i)) \right|$$

As all z_i and z_i' are drawn iid from the distribution $\mathcal{D}$, we can switch them. What happens to

$$\ell(h, z_i) - \ell(h, z_i') ?$$

we get a sign flip in the sum!

We can let $\sigma = (+1, -1, +1, \dots)$ of length m and write

$$\frac{1}{m} \left| \sum_{i=1}^m \sigma_i \cdot (\ell(h, z'_i) - \ell(h, z_i)) \right|$$

(the σ_i 's are called **RADENACHER** variables). We can thus draw σ_i from a uniform distribution on $\{+1, -1\}$, $U(\{\pm 1\})$, take the expectation wrt. σ and still not change the distribution of $\ell(h, z'_i) - \ell(h, z_i)$.

Formally,

$$\begin{aligned} & \mathbb{E}_S \left[\mathbb{1}_{\exists h \in H: |\ell_D(h) - \ell_S(h)| > \varepsilon} \right] \\ & \leq 2 \cdot \mathbb{E}_{S, S'} \left[\mathbb{P}_{\sigma \sim U(\{\pm 1\})} \left[\exists h \in H: \frac{1}{m} \cdot \left| \sum_{i=1}^m \sigma_i \cdot \overbrace{(\ell(h, z'_i) - \ell(h, z_i))}^{\text{range is } [0, c]} \right| > \frac{\varepsilon}{2} \right] \right] \end{aligned}$$

Let's fix S, S' and $h \in H$: let $\theta_h = \frac{1}{m} \cdot \sum_{i=1}^m \sigma_i (\ell(h, z'_i) - \ell(h, z_i))$

We know $\mathbb{E}_{\sigma} [\theta_h] = 0$, since $(\ell(h, z'_i) - \ell(h, z_i))$ is constant (as we fixed h).

This means, we can write

$$\mathbb{P} \left[\left| \theta_h - \underbrace{\mathbb{E}[\theta_h]}_0 \right| > \frac{\varepsilon}{2} \right] \stackrel{\text{Hoeffding}}{\leq} 2 \cdot e^{-2m^2 \left(\frac{\varepsilon}{2}\right)^2 / \sum_i \underbrace{(c - (-c))^2}_{\substack{\text{because the range} \\ \text{of } \theta_h \text{ is } [-c, c]}}}$$

$$= 2 \cdot e^{-m \varepsilon^2 / 8c^2}$$

We can now account for " $\exists h \in H$ " by the union bound. This works as $C = S \cup S'$ contains at most $2m$ instances. So, effectively we have H_C and not H anymore.

06/02/26

So, we now have

$$\mathbb{P}_\sigma \left[\exists h \in H_C : \left| \frac{1}{m} \cdot \sum_{i=1}^m \sigma_i (l(h, z_i') - l(h, z_i)) \right| > \frac{\varepsilon}{2} \right] \stackrel{\text{union bound}}{\leq}$$

$$\underbrace{\gamma_H(2m)}_{\substack{\text{from the growth function, we know how} \\ \text{many labelings are possible, namely } \gamma_H(2m)}} \cdot 2 \cdot e^{-m \varepsilon^2 / 8c^2}$$

from the growth function, we know how many labelings are possible, namely $\gamma_H(2m)$

Overall, we get

$$\mathbb{P}_S \left[\exists h \in H : |L_D(h) - L_S(h)| > \varepsilon \right] \leq 4 \cdot \zeta_H(2m) \cdot e^{-m\varepsilon^2/\rho_c^2}$$

To get the result from the theorem, we set

$$\begin{aligned} 4 \zeta_H(2m) \cdot e^{-m\varepsilon^2/\rho_c^2} &= \delta \\ e^{-m\varepsilon^2/\rho_c^2} &= \frac{\delta}{4 \cdot \zeta_H(2m)} \\ \varepsilon^2 &= \frac{\log\left(\frac{4 \zeta_H(2m)}{\delta}\right) \cdot \rho_c^2}{m} \\ \rightarrow \varepsilon &= \rho_c \cdot \sqrt{\frac{\delta \cdot \log\left(\frac{4 \zeta_H(2m)}{\delta}\right)}{m}} \end{aligned}$$

The assumption that we made, i.e.,

$$m \geq 4c^2 \varepsilon^{-2} \log(2)$$

is equivalent to $\delta \leq 2 \cdot T_2 \cdot \zeta_H(2m)$

Always fulfilled for $\delta \in (0, 1)$

□

We can now formulate the **fundamental theorem of statistical learning**:

Theorem: Let H be a hypothesis class of functions from X to $\{0, 1\}$ and let ℓ be the 0-1 loss. Then, the following statements are equivalent:

- [1] H has the uniform convergence property (UC)
- [2] Any ERM algorithm is a successful agnostic PAC learner for H
- [3] H is agnostic PAC learnable
- [4] H is PAC learnable
- [5] Any ERM algorithm is a successful PAC learner for H
- [6] H has finite VC dimension.

Let's look at some hypothesis classes.

Linear predictors

Define

$$L_d = \left\{ h_{w,b} : w \in \mathbb{R}^d, b \in \mathbb{R} \right\} \left. \vphantom{\begin{matrix} h_{w,b} \\ w \in \mathbb{R}^d, b \in \mathbb{R} \end{matrix}} \right\} \begin{matrix} \text{Class of} \\ \text{affine functions} \end{matrix}$$
$$h_{w,b}(x) = \langle w, x \rangle + b$$

From L_d , we can get different predictors via composition with some $\phi : \mathbb{R} \rightarrow \mathcal{Y}$. Further, we can use

$$w' = (b, w_1, \dots, w_d) \text{ and } x' = (1, x_1, \dots, x_d) \text{ to write}$$

$$h_{w,b}(x) = \langle w', x' \rangle$$

$\Rightarrow$ Affine functions in $\mathbb{R}^d$ can be written as homogeneous linear functions in $\mathbb{R}^{d+1}$.

if $\phi : \mathbb{R} \rightarrow \mathcal{Y}$ is $\phi = \text{sign}$
we get so called HALFSPACE
classifiers HS_d .

$$HS_d = \phi \circ L_d = \left\{ x \mapsto \text{sign}(h_{w,b}(x)), h_{w,b} \in L_d \right\}$$

Question: What is $VC(HS_d)$?

Theorem: The VC dimension of

$$\mathcal{F} = \{ x \mapsto \text{sign}(\langle w, x \rangle) : w \in \mathbb{R}^d \}$$

 is d .

(Remark: $\mathcal{Y} = \{\pm 1\}$)

Proof: We first show $VC(\mathcal{F}) \geq d$ (lower bound)

Let $C = \{c_1, \dots, c_d\}$ where $c_i \in \mathbb{R}^d$

$$c_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, c_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \end{pmatrix}, \dots, c_d = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$$

Let $w = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_d \end{pmatrix}$ for any given labeling (in $\{\pm 1\}$) of $\{c_1, \dots, c_d\}$

$$\Rightarrow \langle w, c_i \rangle = \sum_{j=1}^d w_j \cdot \underbrace{c_{ij}}_{1 \text{ at } i=j} = w_i = y_i$$

j-th coordinate of c_i

1 at $i=j$

$$\Rightarrow \boxed{VC(\mathcal{F}) \geq d}$$

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To establish $VC(\mathcal{F}) = d$, we need to show
 $VC(\mathcal{F}) < d+1$.

We will show this by contradiction.

Assume $d+1$ points $C = \{c_1, \dots, c_{d+1}\}$ are shattered by $\mathcal{F}$.

This means that $\exists w_1, \dots, w_{d+1}$ weight vectors that will yield all 2^{d+1} possible labelings of C .

Lets write down all possible inner products:

$$M = \begin{pmatrix} w_1^T c_1 & w_2^T c_1 & \dots & w_{d+1}^T c_1 \\ w_1^T c_2 & & & \vdots \\ \vdots & & \ddots & \\ w_1^T c_{d+1} & \dots & \dots & w_{d+1}^T c_{d+1} \end{pmatrix} \begin{matrix} \downarrow d+1 \text{ rows} \\ \\ \\ \end{matrix}$$

$\xrightarrow{2^{d+1} \text{ columns}}$

Lets take

$$X = \begin{pmatrix} \text{---} c_1 \text{---} \\ \text{---} c_2 \text{---} \\ \vdots \\ \text{---} c_{d+1} \text{---} \end{pmatrix} \begin{matrix} \xrightarrow{d \text{ cols}} \\ \downarrow d+1 \text{ rows} \end{matrix}$$

and

$$W = \begin{pmatrix} | & | & \dots & | \\ w_1 & w_2 & \dots & w_{2^{d+1}} \\ | & | & \dots & | \end{pmatrix} \begin{matrix} \downarrow d \text{ rows} \\ \\ \end{matrix}$$

$\xrightarrow{\hspace{10em}} 2^{d+1} \text{ columns}$

We can now write

$$M = XW$$

Remark: if we would take $\text{sign}(XW)$, we would get all possible labelings.

We know that

$$\text{rank}(M) \leq \min \{ \text{rank}(X), \text{rank}(W) \} = d$$

$$M = \begin{pmatrix} w_1^T c_1 & & & w_{2d+1}^T c_1 \\ w_1^T c_2 & & & \vdots \\ \vdots & & & \vdots \\ w_1^T c_{d+1} & \dots & \dots & w_{2d+1}^T c_{d+1} \end{pmatrix}$$

$\chi w_i \in \mathbb{R}^{d+1}$

Any $v_1, \dots, v_k$ (set of vectors) are linearly independent if and only if (x)

$$a_1 \cdot v_1 + a_2 \cdot v_2 + \dots + a_k v_k = 0$$

for $\forall i: a_i = 0$

In our case (for M), if we look at the rows of M

$$\begin{aligned} & a_1 \cdot [w_1^T c_1 \dots \dots \dots w_{2d+1}^T c_1] + \\ & a_2 \cdot [w_1^T c_2 \dots \dots \dots w_{2d+1}^T c_2] + \\ & \vdots \\ & a_{d+1} \cdot [w_1^T c_{d+1} \dots \dots \dots w_{2d+1}^T c_{d+1}] \end{aligned}$$

letting $a = [a_1, \dots, a_{d+1}]$, we can write this as

$$[q^T X w_1 \quad q^T X w_2 \quad \dots \quad q^T X w_{2d+1}]$$

(by assumption)

As we assume that $c_1, \dots, c_{d+1}$ are shattered by $\mathcal{F}$, there

$\exists k \in \{1, \dots, 2^{d+1}\}$ such that

$$\text{sign}(q) = \text{sign}(X w_k)$$

This means that

$$q^T X w_k$$

is a sum of positive numbers $\Rightarrow \neq 0$! In turn, by (x) this means we have $d+1$ linearly independent rows in M , i.e., $\text{rank}(M) = d+1$.

$\Rightarrow$ contradiction to our earlier statement of $\text{rank}(M) \leq d$!

We conclude that $VC(\mathcal{F}) < d+1$ and thus (with our lower bound)

$$\boxed{VC(\mathcal{F}) = d}$$

□

Boosting

Def. (ρ -weak learnability): A learning algorithm A is a ρ -weak learner for a hypothesis class H if there exists $m_H : (0, 1) \rightarrow \mathbb{N}$ such that for every $\delta \in (0, 1)$, for every distribution D over X , and for every labeling function $f : X \rightarrow \{\pm 1\}$, if realizability holds with respect to H, D, f , then when running A on $m \geq m_H(\delta)$ iid samples from D (labeled by f), the algorithm returns a hypothesis h such that with probability of at least $1 - \delta$ (over the choice of m samples from D)

$$L_{D,f}(h) \leq \frac{1}{2} - \rho$$

Requirements : Base learner B

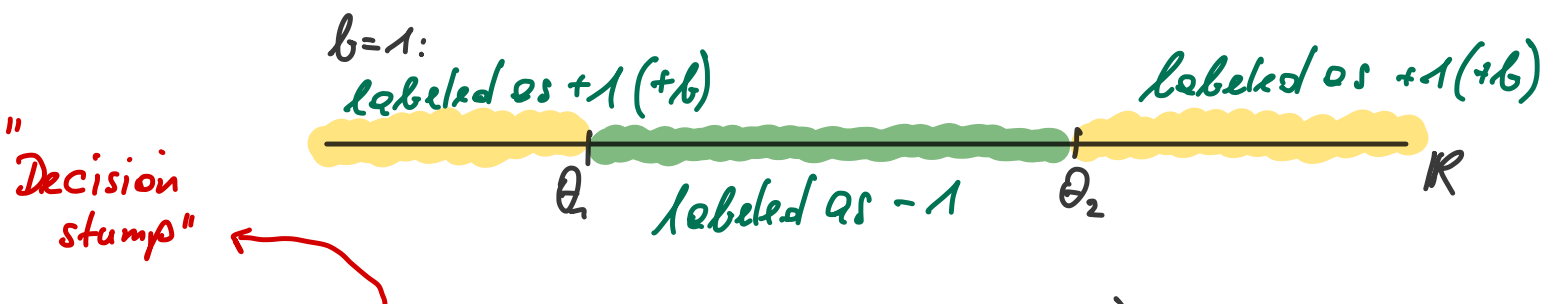
[1] ERH_B is efficiently implementable

[2] For every sample S , labeled by $h \in H$, any ERH_B hypothesis has an error of at most $\frac{1}{2} - \rho$

Example: $X = \mathbb{R}$

$$H = \{h_{\theta_1, \theta_2, b} : \theta_1, \theta_2 \in \mathbb{R}, b \in \{\pm 1\}, \theta_1 < \theta_2\}$$

$$h_{\theta_1, \theta_2, b}(x) = \begin{cases} +b, & \text{if } x < \theta_1 \text{ or } x > \theta_2 \\ -b, & \text{else} \end{cases}$$



$$\text{Base class } \mathcal{B} = \{x \mapsto \text{sign}(x - \theta) \cdot b, \theta \in \mathbb{R}, b \in \{\pm 1\}\}$$

Claim: $\mathcal{B}$ is a $\frac{1}{12}$ -weak learner for H .

Proof:

1 For every distribution $\mathcal{D}$ consistent with H , there exists a decision stump that agrees with the labeling of any pair of components from the 3-part partitioning of $\mathbb{R}$ by some $h \in H$!

2 For any partitioning of $\mathbb{R}$ into 3 parts (by h), one of these regions/parts must have a probability mass of $\leq \frac{1}{3}$.

1, 2 $\Rightarrow$ given a zero-error hypothesis $h \in H$, any decision stump that disagrees with h on such a region (from 2) has error $\leq \frac{1}{3}$.

The VC-Dimension of B is 2 $\Rightarrow$ running ERM_B on $m \geq m_B(\epsilon, \delta)$ samples, we have with probability of at least $1-\delta$ that the error is at most

$$\frac{1}{3} + \epsilon$$

lets set $\epsilon = \frac{1}{12} \Rightarrow L_{D,f}(h) \leq \frac{1}{2} - \frac{1}{12}$ ↙
 $\text{ERM hyp. from } B!!!$

ADABOOST (Adaptive Boosting)

Idea: Use a "weak learner" to build a "strong learner".

Algorithm:

Input: $(x_1, y_1), \dots, (x_m, y_m)$

// $y_i \in \{\pm 1\}$

for $i=1, \dots, m$
 $D_1(i) = \frac{1}{m}$
 end

for $t=1, \dots, T$ // $T \dots$ nr. of boosting rounds

$h_t \leftarrow$ base classifier with error

$$\epsilon_t = \sum_{i \sim D_t} [h_t(x_i) \neq y_i]$$

$$\alpha_t \leftarrow \frac{1}{2} \cdot \log \left(\frac{1 - \epsilon_t}{\epsilon_t} \right)$$

$$z_t \leftarrow 2 \cdot [\epsilon_t \cdot (1 - \epsilon_t)]^{\frac{1}{2}}$$

```

:   |   for  $i=1, \dots, m$ 
:   |    $D_{t+1}(i) = \frac{1}{Z_t} \cdot D_t(i) \cdot e^{-y_i \alpha_t h_t(x_i)}$  (X)
:   |   end
:   end
g ←  $\sum_{t=1}^T \alpha_t \cdot h_t$ 
(return sign( $\sum_{t=1}^T \alpha_t h_t$ ))

```

Observation 1: if $\epsilon_t < \frac{1}{2}$, then $\frac{1-\epsilon_t}{\epsilon_t} > 1$ and thus $\alpha_t > 0$

This means: $\underbrace{\alpha_t}_{>0} \cdot \underbrace{y_i \cdot h_t(x_i)}_{\in \{\pm 1\}}$

Case 1: $y_i = 1, h_t(x_i) = -1$

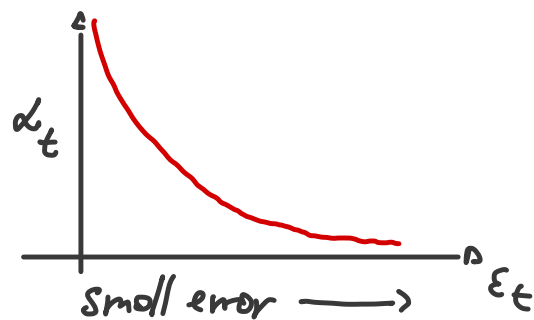
$e^{-y_i \alpha_t h_t(x_i)}$ is going to be large.

Case 2: $y_i = -1, h_t(x_i) = 1$
 $\Rightarrow$ same situation

Hence, if the i -th sample is incorrectly classified, its weight in (X) increases.

Observation 2:

$$\alpha_t = \frac{1}{2} \cdot \log \left(\frac{1 - \epsilon_t}{\epsilon_t} \right)$$



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For $t \in \{1, \dots, T\}$, we let f_t denote the combination of base classifiers up to round t . We have $f = f_T$.

Also, we have

$$\forall i \in \{1, \dots, m\}: D_{t+1}(i) = \frac{e^{-y_i f_t(x_i)}}{m \cdot \prod_{s=1, \dots, t} z_s}$$

Why? In the algorithm, we had

$$D_{t+1}(i) = \frac{1}{z_t} \cdot D_t(i) \cdot e^{-y_i \alpha_t h_t(x_i)}$$

$$(D_1(i) = \frac{1}{m})$$

$$= \frac{D_{t-1}(i) \cdot e^{-y_i \alpha_{t-1} h_{t-1}(x_i)} \cdot e^{-y_i \alpha_t h_t(x_i)}}{z_t \cdot z_{t-1}}$$

$$\vdots$$
$$= \frac{e^{-y_i \sum_{s=1}^t \alpha_s \cdot h_s(x_i)}}{m \cdot \prod_{s=1, \dots, t} z_s}$$

Noting that $f_t = \sum_{s=1}^t \alpha_s \cdot h_s$, we get that

$$D_{t+1}(i) = \frac{e^{-y_i f_t(x_i)}}{m \cdot \sum_{s=1}^t z_s} \quad (**)$$

$\swarrow$
i-th training instance

Theorem: Let S be a training set and assume that at each iteration of ADABOOST, the weak learner returns a hypothesis with error of at most $\frac{1}{2} - \rho$. Then

$$L_S(h_S) \leq e^{-2\rho^2 T}$$

↪ returned by ADABOOST after T rounds.

More generally,

$$L_S(h_S) \leq e^{-2 \cdot \sum_{t=1}^T \left(\frac{1}{2} - \epsilon_t\right)^2}$$

Proof: $\mathbb{1}_{u \leq 0} \leq e^{-u}$ for all $u \in \mathbb{R}$ (xxx)

we can write

$$L_S(h) = \frac{1}{m} \cdot \sum_{i=1}^m \mathbb{1}_{\underbrace{y_i p_T(x_i) \leq 0}_{\text{this is when we make a mistake}}}$$

$$\stackrel{\text{by (xxx)}}{\leq} \frac{1}{m} \cdot \sum_{i=1}^m e^{-y_i p_T(x_i)}$$

$$= \frac{1}{m} \cdot \sum_{i=1}^m \left[m \cdot \frac{1}{\|z_t\|} \right] \cdot D_{T+1}(i)$$

$$= \frac{1}{\|z_t\|} \underbrace{\sum_{i=1}^m D_{T+1}(i)}_1$$

$$= \frac{1}{\|z_t\|}$$

we know that z_t is a normalization term in

$$D_{t+1}(i) = \frac{D_t(i) \cdot e^{-y_i d_t h_t(x_i)}}{\sum_{j=1}^m D_t(j) \cdot e^{-y_j d_t h_t(x_j)}} \quad (\text{see Algorithm})$$

↪ this is our z_t

Next, we are going to use this expression for z_t and try to split it up.

$$\sum_{i=1}^m D_t(i) \cdot e^{-y_i \alpha_t h_t(x_i)} = \sum_{i: y_i h_t(x_i) = +1} D_t(i) \cdot e^{-\alpha_t} + \sum_{i: y_i h_t(x_i) = -1} D_t(i) \cdot e^{\alpha_t}$$

Let's look at

$$\sum_{i: y_i h_t(x_i) = +1} D_t(i) e^{-\alpha_t}$$

in more detail.

$$\Rightarrow e^{-\alpha_t} \cdot \sum_{i: y_i h_t(x_i) = +1} D_t(i)$$

We also know

$$\varepsilon_t = \sum_{i=1}^m D_t(i) \cdot \mathbb{1}_{y_i \neq h_t(x_i)}$$

Consequently, we have

$$\sum_{i: y_i h_t(x_i) = +1} D_t(i) \cdot e^{-\alpha_t} = e^{-\alpha_t} \cdot \sum_{i: y_i h_t(x_i) = +1} D_t(i) = e^{-\alpha_t} \cdot (1 - \varepsilon_t)$$

Overall, we have

$$\sum_{i=1}^m D_t(i) \cdot e^{-y_i \alpha_t h_t(x_i)} = e^{-\alpha_t} \cdot (1 - \varepsilon_t) + e^{\alpha_t} \cdot \varepsilon_t$$

Lets set in α_t from the ADABOOST algorithm:

$$\begin{aligned} & e^{-\left(\frac{1}{2} \cdot \log\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right)\right)} \cdot (1 - \varepsilon_t) + e^{\left(\frac{1}{2} \cdot \log\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right)\right)} \cdot \varepsilon_t \\ &= (1 - \varepsilon_t) \cdot \sqrt{\frac{\varepsilon_t}{1 - \varepsilon_t}} + \varepsilon_t \cdot \sqrt{\frac{1 - \varepsilon_t}{\varepsilon_t}} \\ &= 2 \cdot \sqrt{\varepsilon_t \cdot (1 - \varepsilon_t)} \end{aligned}$$

We know have a convenient expression for z_t and thus

$$L_S(h) \leq \sum_{t=1}^T z_t = \sum_{t=1}^T 2 \cdot \sqrt{\varepsilon_t \cdot (1 - \varepsilon_t)}$$

Remark: $\sqrt{4\varepsilon_t \cdot (1 - \varepsilon_t)} = \sqrt{4\varepsilon_t - 4\varepsilon_t^2}$

$$\begin{aligned} &= \sqrt{4\varepsilon_t - 4\varepsilon_t^2 - 1 + 1} \\ &= \sqrt{1 - 4 \cdot \underbrace{\left(\frac{1}{4} - \varepsilon_t + \varepsilon_t^2\right)}_{\left(\frac{1}{2} - \varepsilon_t\right)^2}} = \sqrt{1 - 4 \cdot \left(\frac{1}{2} - \varepsilon_t\right)^2} \end{aligned}$$

$$\begin{aligned}
L_S(h) &\leq \prod_{t=1}^T 2 \cdot \sqrt{\varepsilon_t \cdot (1 - \varepsilon_t)} \\
&= \prod_{t=1}^T \sqrt{1 - 4 \cdot \left(\frac{1}{2} - \varepsilon_t\right)^2} \quad (\text{from Remark}) \\
&\leq \prod_{t=1}^T \sqrt{e^{-4 \cdot \left(\frac{1}{2} - \varepsilon_t\right)^2}} \quad (\text{by } 1-x \leq e^{-x}) \\
&= \prod_{t=1}^T e^{-2 \left(\frac{1}{2} - \varepsilon_t\right)^2} \\
&= e^{-2 \cdot \sum_{t=1}^T \left(\frac{1}{2} - \varepsilon_t\right)^2}
\end{aligned}$$

Setting in $\varepsilon_t \leq \frac{1}{2} - \nu$ for $t \in \{1, \dots, T\}$, we get

$$L_S(h) \leq e^{-2\nu^2 T}$$

$$\sum_{t=1}^T \alpha_t \cdot h_t$$

□